Minimal Reviewable Evidence Specification for Constitutional AI

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Abstract

This document specifies a minimal reviewable evidence surface for AI systems claiming constitutional operation under finite capacity. It does not define a full audit standard, implementation protocol, or legal compliance framework. Its narrower purpose is to state the minimum review artefacts required for a constitutional claim to be inspectable rather than merely asserted. The specification defines the evidence objects, their minimum fields, integrity requirements, scope declarations, certificate discipline, and non-claim boundaries. It is intended to function as a companion object to a constitutional article draft and one or more application profiles.

1 Status of this document

This document is a **minimal evidence specification**. It does not prove that any AI system is safe, ethical, lawful, unbiased, or fit for deployment in a specific domain. It specifies only the minimum evidence conditions under which a constitutional claim may become reviewable.

Its function is therefore narrow:

> to prevent constitutional language from floating free of inspectable evidence.

2 Purpose

A constitutional AI claim must not remain a slogan. If a system claims that it:

- distinguishes admissibility from execution,
- can hold or refuse lawfully,
- preserves witness,
- remains reviewable under finite capacity,

then a minimum evidence layer must exist through which those claims may be inspected.

This specification defines that minimum layer.

3 Scope

This specification applies to AI systems that make claims such as:

- constitutional operation,
- reviewable refusal,
- explicit holding,
- witness-preserving boundary governance,
- bounded admissibility discipline,
- or certificate-bearing accountable operation.

It may be used together with:

- a constitutional article draft,
- an application profile,
- a domain-specific deployment profile,
- or a runtime demonstration pack.

4 Evidence principle

The core evidence principle is:

> No constitutional act without witness; no witness without scope; no scope without review conditions.

Accordingly, a minimal reviewable evidence package must always make visible:

- **what** act occurred,
- **why** it was declared,
- **under which system and profile** it occurred,
- **what the evidence does and does not support**,
- and **how integrity is preserved**.

5 Minimal evidence package

A minimal reviewable evidence package should contain, at minimum, the following artefacts:

1. a `manifest.json`
2. an action trace such as `actions.jsonl`
3. optionally, a frame or state trace such as `frames.jsonl`
4. an integrity file such as `checksums.sha256`

5. a short natural-language scope note such as `README.md`
6. and, where a verdict is issued, a `certificate.json` or certificate envelope

These filenames are illustrative rather than mandatory. What is mandatory is the review function they serve.

6 Required artefact functions

6.1 Manifest

The manifest identifies the evidence package.

It should minimally declare:

- system name,
- profile name,
- version,
- date or timestamp,
- scenario or run label,
- applicable domain profile,
- relevant configuration scope,
- and package contents.

6.2 Action trace

The action trace records constitutionally significant acts.

It should minimally allow a reviewer to inspect:

- event order,
- the declared act,
- the associated reason label or reason family,
- the interaction or step identifier,
- and any scoped witness object or pointer attached to the act.

6.3 Frame or state trace

Where needed, the frame trace records the surrounding runtime surface within which an act occurred.

It may include:

- local telemetry,

- bounded pressure indicators,
- tool state,
- review-relevant flags,
- or other declared witness fields.

It is not mandatory in all settings, but where omitted, the package must still remain reviewable through the available artefacts.

6.4 Integrity file

The integrity file allows the reviewer to verify that the package has not been silently altered.

It should minimally support file-level integrity checking across the included artefacts.

6.5 Scope note

The scope note explains:

- what is being claimed,
- what is not being claimed,
- what profile or scenario the package belongs to,
- what limitations govern interpretation,
- and what a reviewer may legitimately conclude from the package.

6.6 Certificate object

Where a certificate is issued, it must remain subordinate to the evidence.

A certificate must not replace the evidence package. It may summarise, but it must not erase ambiguity, scope limits, or lower-layer qualifications.

7 Minimum fields

7.1 Minimum fields for `manifest.json`

A minimal manifest should include fields equivalent to the following:

field	function
system_name	identifies the system under review
profile_name	identifies the relevant application or domain profile
version	identifies the system/profile version
run_label	identifies the specific run or evidence instance
timestamp_utc	anchors the package in time
package_contents	lists included artefacts
scope	states what this package is meant to evidence
non_claims	states what this package does not establish
integrity_method	identifies how integrity is checked

7.2 Minimum fields for an action trace

Each constitutionally significant line or event should minimally expose fields equivalent to:

field	function
step / event_id	preserves event order
timestamp_utc	anchors the event in time
action_type	records the declared act
reason_code	records why the act was declared
witness	attaches a bounded review object or pointer
profile_version	ties the act to a declared profile/version

7.3 Minimum fields for a certificate

If used, a certificate should minimally expose fields equivalent to:

field	function
certificate_type	states the type of verdict object
scope	states what is being certified
status	states the verdict or classification
qualifications	states lower-layer limits or caveats
evidence_refs	points back to the supporting evidence
non_claims	prevents over-interpretation

8 Act evidence requirements

8.1 ACCEPT

An ACCEPT event should be reviewable as an explicit admission of continuation.

Minimal review requirement:

- the act must be explicit,
- it must not be inferred merely from the presence of output,

- and it must remain tied to a declared profile and reasonable scope.

8.2 `HOLD`

A `HOLD` event must not appear as passive delay, silent wait, or hidden buffering.

Minimal review requirement:

- the act must be explicit,
- it must carry a reason family,
- and the package must make visible that lawful continuation was deferred rather than silently abandoned.

8.3 `REFUSE`

A `REFUSE` event must not be substituted by timeout, unexplained absence, or opaque failure.

Minimal review requirement:

- the refusal must be explicit,
- it must carry a reason family,
- and it must remain inspectable as a constitutional act rather than a transport or infrastructure accident.

8.4 Extension acts

If the system declares extension acts such as `ESCALATE`, `TRANSFER`, or `RE-DESCRIBE`, the evidence package must make visible:

- that the extension act occurred,
- under what declared scope it is valid,
- and how responsibility and witness are preserved across the transition.

9 Reason-label discipline

Reason labels shall be:

- declared,
- scoped,
- semantically stable across a package or profile,
- and sufficient for later review.

Reason labels need not expose raw thresholds, coefficients, or proprietary mechanisms. However, they must not be so vague that the act becomes unreviewable.

Illustrative reason families include:

- `OVERLOAD`
- `UNDER_REVIEW`
- `SAFETY_BOUNDARY`
- `SCOPE_LIMIT`
- `INSUFFICIENT_BASIS`
- `ESCALATION_REQUIRED`

10 Witness discipline

A witness object may be minimal, but it must not be empty.

At minimum, the witness must help the reviewer answer:

> Why should this event count as a constitutionally meaningful act under the declared profile?

A witness may be:

- inline in the action trace,
- attached as a bounded object,
- or referenced via a linked artefact,

provided that it remains accessible and scoped.

11 Scope and non-claim discipline

Every evidence package must explicitly distinguish:

- **what it supports**,
- **what it illustrates**,
- **what it does not prove**,
- and **under what profile or scenario it is valid**.

This is necessary to prevent:

- witness inflation,
- certificate overreach,
- demo-to-universal drift,
- and public-facing overclaim.

12 Qualification inheritance

No certificate or higher-level summary may erase unresolved lower-layer weakness.

Therefore:

- if the evidence is scenario-bound, the certificate must say so;
- if the witness is partial, the certificate must say so;
- if lower-layer instability remains unresolved, the higher verdict must inherit that qualification explicitly.

A reviewer must never be asked to treat a high-level verdict as stronger than the evidence beneath it.

13 Protected non-disclosure

This specification does not require full exposure of:

- raw thresholds,
- model weights,
- proprietary parameters,
- or security-sensitive internals.

But if such material is withheld, the package must still preserve sufficient witness for meaningful review.

Non-disclosure is constitutionally valid only if reviewability survives.

14 Minimal package example

A minimal reviewable evidence package may therefore look like:

```
/
|- manifest.json
|- actions.jsonl
|- frames.jsonl           (optional but recommended where relevant)
|- checksums.sha256
|- certificate.json       (optional, if a verdict is issued)
|- README.md
```

The exact filenames may vary. The constitutional requirement lies in the review function, not in the filename itself.

15 Failure modes this specification is meant to prevent

This specification is designed to reduce the risk of:

- silent degradation,
- timeout-as-refusal,
- unscoped refusal,
- witness-free continuation,
- certificate theatre,
- opaque escalation,
- non-reviewable holding,
- and scope drift between evidence and public claim.

16 Non-claims

A package satisfying this specification does *not* by itself prove:

- ethical legitimacy,
- legal compliance,
- universal safety,
- fairness,
- domain fitness,
- or deployment readiness.

It proves something narrower:

> that the constitutional claim being made has been attached to a minimum reviewable evidence surface.

17 Minimal conformance checklist

A package minimally conforms to this specification only if all of the following are true:

1. It identifies the system, profile, version, and evidence instance.
2. It preserves an explicit action trace for constitutionally significant acts.
3. It includes a manifest stating scope and non-claims.
4. It includes an integrity mechanism.
5. It provides reason-label discipline for non-trivial acts.
6. It preserves bounded witness for later review.
7. It distinguishes evidence from certificate.
8. It preserves qualification inheritance across levels.

9. It does not allow non-disclosure to erase reviewability.
10. It does not present the package as stronger than its declared scope.

18 Closing statement

A constitutional claim without evidence is rhetoric. Evidence without scope is ambiguity. A certificate without qualification is overreach.

This specification therefore establishes a minimal rule: if an AI system claims constitutional operation under finite capacity, it must leave behind a reviewable evidence surface through which its boundary acts can be inspected. Without that surface, constitutional language remains structurally weak.